

Core Pure Mathematics 2

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Chapter 4: Volumes of Revolution — Further

Why this matters

A Level Maths covers volumes of revolution about the x-axis for a curve $y = f(x)$. Further Maths extends this in two directions: rotating about the **y-axis**, and rotating a curve given **parametrically**. Both are natural extensions of the same core idea (summing infinitesimally thin discs), but each needs a small but important adjustment to the setup.

Concept: volume of revolution about the y-axis

When a region is rotated 2π radians about the y-axis, we build the solid from thin discs stacked along the y-axis, each with radius x and thickness dy . This means the curve must be expressed (or thought of) as x in terms of y :

$$\text{Volume (about y-axis)} = \pi \int_c^d x^2 dy$$

where c and d are the y-values bounding the region. If the curve is given as $y = f(x)$, you may need to rearrange to $x = g(y)$ first, or substitute using the chain rule if rearranging is awkward.

Key formula

KEY FORMULA

Rotation about	Formula	Variable of integration
x-axis	$V = \pi \int_a^b y^2 dx$	x
y-axis	$V = \pi \int_c^d x^2 dy$	y

Worked example 1 — Volume of revolution about the y-axis

WORKED EXAMPLE 1

Question. The region bounded by the curve $y = x^2$, the y-axis, and the lines $y = 1$ and $y = 4$ is rotated 2π radians about the y-axis. Find the exact volume generated.

Step 1. Rearrange $y = x^2$ to give x^2 in terms of y : since $y = x^2$, we have $x^2 = y$ directly. **(M1)**

Step 2. Set up the volume integral using the y-axis formula with limits $y = 1$ to $y = 4$: **(M1)** $V = \pi \int_1^4 x^2 dy = \pi \int_1^4 y dy$

Step 3. Integrate: **(M1)** $\pi \int_1^4 y dy = \pi [y^2/2]$ from 1 to 4 $= \pi(16/2 - 1/2) = \pi(8 - 0.5) = 7.5\pi$

Answer: $V = 7.5\pi$ (or $15\pi/2$) cubic units. **(A1)**

EXAM TIP

The rearrangement step (x^2 in terms of y) is where most errors happen — either forgetting to square x before substituting, or trying to find $x = \sqrt{y}$ and then square it again unnecessarily, introducing sign or domain errors. Remember that you only ever need x^2 , never x itself, so if a curve is given as $y = f(x)$ and x^2 pops out directly (as here, since $y = x^2$ means $x^2 = y$), don't over-complicate by solving for x first.

Concept: volume of revolution for a parametric curve

When a curve is defined parametrically by $x = x(t)$, $y = y(t)$, we convert the volume integral using the substitution $dx = (dx/dt) dt$ (for rotation about the x -axis) or $dy = (dy/dt) dt$ (for rotation about the y -axis), and change the limits from x - or y -values to the corresponding t -values.

This is just a change of variable in the integral, exactly like integration by substitution: since x and y are both functions of t , the disc-slice thickness dx is really $(dx/dt) \cdot dt$, so replacing dx with $(dx/dt) dt$ and swapping the x -limits for the matching t -limits turns the ordinary volume integral into one you can evaluate directly in t — no need to ever find y as an explicit function of x first.

Volume about x -axis (parametric): $V = \pi \int(t_1 \text{ to } t_2) y^2 (dx/dt) dt$

Volume about y -axis (parametric): $V = \pi \int(t_1 \text{ to } t_2) x^2 (dy/dt) dt$

Always check the direction of t as it increases — if x decreases as t increases, you may need to swap the limits (or take the negative) to get a positive volume.

Worked example 2 — Volume of revolution for a parametric curve, about the x -axis

WORKED EXAMPLE 2

Question. A curve is defined parametrically by $x = t^2$, $y = 2t$, for $t \geq 0$. Find the volume generated when the region bounded by the curve, the x -axis, and the line $x = 4$ is rotated 2π radians about the x -axis.

Step 1. Find the t -values corresponding to the limits: $x = 0$ gives $t = 0$; $x = 4$ gives $t^2 = 4$, so $t = 2$ (taking $t \geq 0$). (M1)

Step 2. Find dx/dt : $x = t^2$, so $dx/dt = 2t$. (M1)

Step 3. Set up the parametric volume integral: (M1) $V = \pi \int_0^2 y^2 (dx/dt) dt = \pi \int_0^2 (2t)^2 (2t) dt = \pi \int_0^2 4t^2 \cdot 2t dt = \pi \int_0^2 8t^3 dt$

Step 4. Integrate: (A1) $\pi \int_0^2 8t^3 dt = \pi [2t^4]$ from 0 to 2 $= \pi(2 \cdot 16 - 0) = 32\pi$

Answer: $V = 32\pi$ cubic units. (A1)

EXAM TIP

A common slip is substituting y^2 and dx/dt but forgetting to also change the *limits* from x -values to t -values — integrating $y^2(dx/dt)$ from $x = 0$ to $x = 4$ instead of from $t = 0$ to $t = 2$, which is meaningless (mixing variables). Always write "when $x = 0$, $t = \dots$ " and "when $x = 4$, $t = \dots$ " explicitly as a first line before setting up the integral, exactly as with parametric arc length or area problems.

Worked example 3 — Volume of revolution for a parametric curve, about the y-axis

WORKED EXAMPLE 3

Question. A curve is defined parametrically by $x = 3\cos \theta$, $y = 3\sin \theta$, for $0 \leq \theta \leq \pi/2$. Find the volume generated when the region bounded by the curve and the coordinate axes is rotated 2π radians about the y-axis.

Step 1. Find the θ -values at the limits: at $y = 0$, $\sin \theta = 0$ so $\theta = 0$; at $y = 3$ (the top of the curve, $x = 0$), $\sin \theta = 1$ so $\theta = \pi/2$. **(M1)**

Step 2. Find $dy/d\theta$: $y = 3\sin \theta$, so $dy/d\theta = 3\cos \theta$. **(M1)**

Step 3. Set up the parametric volume integral about the y-axis: **(M1)** $V = \pi \int_{(\theta=0 \text{ to } \pi/2)} x^2 (dy/d\theta) d\theta = \pi \int_0^{\pi/2} (3\cos \theta)^2 (3\cos \theta) d\theta = \pi \int_0^{\pi/2} 27\cos^3 \theta d\theta$

Step 4. Integrate $\cos^3 \theta$ using $\cos^3 \theta = \cos \theta(1 - \sin^2 \theta)$: **(M1)** — **standard trig integration technique from Core Pure/Core Maths** $\int \cos^3 \theta d\theta = \int \cos \theta d\theta - \int \sin^2 \theta \cos \theta d\theta = \sin \theta - \sin^3 \theta / 3 + c$

Step 5. Evaluate: **(A1)** $27\pi [\sin \theta - \sin^3 \theta / 3]$ from 0 to $\pi/2 = 27\pi[(1 - 1/3) - (0 - 0)] = 27\pi(2/3) = 18\pi$

Answer: $V = 18\pi$ cubic units.

Sanity check: $x = 3\cos \theta$, $y = 3\sin \theta$ for $0 \leq \theta \leq \pi/2$ traces a quarter-circle of radius 3, so the region bounded by the curve and the axes is a quarter-disc. Rotating a quarter-disc of radius 3 about the y-axis sweeps out a hemisphere of radius 3, with known volume $(2/3)\pi r^3 = (2/3)\pi(27) = 18\pi$ — exactly matching the integral above. When a parametric curve is secretly a circle or ellipse arc, checking against a known solid (sphere, cone, hemisphere) is a fast way to confirm your answer.

Chapter summary

Idea	Key point
Rotation about y-axis	$V = \pi \int x^2 dy$; requires x^2 as a function of y , or a substitution
Rotation about x-axis (parametric)	$V = \pi \int y^2 (dx/dt) dt$, with limits converted to t -values
Rotation about y-axis (parametric)	$V = \pi \int x^2 (dy/dt) dt$, with limits converted to t -values
Limits	Always convert limits to match the variable of integration before evaluating

Common exam mistakes

- Forgetting to convert the limits of integration from x (or y) values to t -values in parametric problems.
- Using dx/dt when the rotation is about the y-axis (should be dy/dt), or vice versa.

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- Forgetting to square x or y before multiplying by the derivative term.
 - Getting a negative volume from the integral (usually because the limits are the wrong way round, or t decreases as x/y increases) and not spotting that the magnitude, not the signed value, is the required volume.
 - Attempting to convert a parametric curve to Cartesian form when it is unnecessary (and often much harder) — the parametric volume formula avoids this entirely.

You should now be able to

- Find the volume generated when a region is rotated about the y -axis, rearranging the equation of the curve where necessary.
- Find the volume generated when a parametrically-defined curve is rotated about the x -axis or the y -axis.
- Correctly convert limits of integration from x - or y -values into the corresponding parameter values.
- Recognise and correct a negative volume result arising from limits or parameter direction.